

Trigonometric Functions & the Unit Circle

ANSWER KEY



Radians, degrees, and the unit circle

- 1 Convert 150° to radians, and convert $\frac{5\pi}{4}$ radians to degrees.
 $150^\circ \cdot \frac{\pi}{180^\circ} = \frac{5\pi}{6}$ radians. $\frac{5\pi}{4} \cdot \frac{180^\circ}{\pi} = 225^\circ$.
- 2 Find the exact coordinates of the point on the unit circle at angle $\theta = \frac{2\pi}{3}$.
 $\frac{2\pi}{3} = 120^\circ$, a reference angle of 60° in Quadrant II. Coordinates: $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$.

Exact values from the unit circle

- 3 Find the exact values of $\sin\left(\frac{7\pi}{6}\right)$ and $\cos\left(\frac{7\pi}{6}\right)$.
 $\frac{7\pi}{6} = 210^\circ$, Quadrant III, reference angle 30° . $\sin\left(\frac{7\pi}{6}\right) = -\frac{1}{2}$, $\cos\left(\frac{7\pi}{6}\right) = -\frac{\sqrt{3}}{2}$.
- 4 Find the exact value of $\tan\left(\frac{5\pi}{3}\right)$.
 $\frac{5\pi}{3} = 300^\circ$, Quadrant IV, reference angle 60° . $\sin = -\frac{\sqrt{3}}{2}$, $\cos = \frac{1}{2}$, so $\tan\left(\frac{5\pi}{3}\right) = \frac{-\sqrt{3}/2}{1/2} = -\sqrt{3}$.

Visualizing sine and cosine graphs

- 5 The graph shows $f(x) = \sin(x)$ and $g(x) = \cos(x)$ over one period.
 - a State the amplitude and period of each function shown. Both curves have amplitude 1 and period 2π , matching the single full up-down cycle shown across the interval.
 - b Identify the horizontal shift that maps $\sin(x)$ onto $\cos(x)$. $\cos(x) = \sin\left(x + \frac{\pi}{2}\right)$, so shifting $\sin(x)$ left by $\frac{\pi}{2}$ produces $\cos(x)$ — visible as the cosine curve reaching its peak a quarter-period earlier.

Amplitude, period, and vertical shift

- 6 State the amplitude, period, and vertical shift of $f(x) = 3\sin(2x) + 1$.
Amplitude = $|3| = 3$. Period = $\frac{2\pi}{2} = \pi$. Vertical shift = 1 (midline $y = 1$).
- 7 A Ferris wheel's height above the ground is modeled by $h(t) = 20\cos\left(\frac{\pi}{6}t\right) + 22$, where t is in minutes. Find the maximum height, minimum height, and period of one full rotation.
Maximum height = $22 + 20 = 42$ m. Minimum height = $22 - 20 = 2$ m. Period = $\frac{2\pi}{\pi/6} = 12$ minutes.

Reciprocal and Pythagorean identities

- 8 Given $\sin \theta = \frac{3}{5}$ with θ in Quadrant II, find $\cos \theta$ and $\tan \theta$.

Using $\sin^2 \theta + \cos^2 \theta = 1$: $\cos^2 \theta = 1 - \frac{9}{25} = \frac{16}{25}$, so $\cos \theta = \pm \frac{4}{5}$. In Quadrant II, cosine is negative:
 $\cos \theta = -\frac{4}{5}$. $\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{3/5}{-4/5} = -\frac{3}{4}$.

- 9 Find $\sec \theta$ and $\csc \theta$ if $\cos \theta = -\frac{5}{13}$ and θ is in Quadrant III.

$\sec \theta = \frac{1}{\cos \theta} = -\frac{13}{5}$. In Quadrant III, $\sin \theta < 0$: using $\sin^2 \theta = 1 - \frac{25}{169} = \frac{144}{169}$, $\sin \theta = -\frac{12}{13}$, so
 $\csc \theta = -\frac{13}{12}$.

Modeling periodic phenomena

- 10 The average daily temperature in a city is modeled by $T(m) = 15\sin\left(\frac{\pi}{6}(m-3)\right) + 18$ degrees Celsius, where m is the month number ($m = 1$ for January). Find the temperature in April ($m = 4$) and state the month of maximum temperature.

$T(4) = 15\sin\left(\frac{\pi}{6}(1)\right) + 18 = 15\sin\left(\frac{\pi}{6}\right) + 18 = 15(0.5) + 18 = 25.5^\circ \text{C}$. Maximum occurs when $\sin\left(\frac{\pi}{6}(m-3)\right) = 1$, i.e. $\frac{\pi}{6}(m-3) = \frac{\pi}{2}$, giving $m-3 = 3$, so $m = 6$ (June).

Co-function and even/odd identities

- 11 Use a co-function identity to rewrite $\cos\left(\frac{\pi}{5}\right)$ in terms of sine.

$\cos \theta = \sin\left(\frac{\pi}{2} - \theta\right)$, so $\cos\left(\frac{\pi}{5}\right) = \sin\left(\frac{\pi}{2} - \frac{\pi}{5}\right) = \sin\left(\frac{3\pi}{10}\right)$.

- 12 Use the fact that sine is an odd function and cosine is an even function to evaluate $\sin(-30^\circ)$ and $\cos(-60^\circ)$.

Since $\sin(-\theta) = -\sin(\theta)$: $\sin(-30^\circ) = -\sin(30^\circ) = -\frac{1}{2}$. Since $\cos(-\theta) = \cos(\theta)$:
 $\cos(-60^\circ) = \cos(60^\circ) = \frac{1}{2}$.

Solving basic trigonometric equations

- 13 Solve $2\sin \theta - 1 = 0$ for θ in $[0, 2\pi)$.

$\sin \theta = \frac{1}{2}$. Reference angle $\frac{\pi}{6}$, with sine positive in Quadrants I and II: $\theta = \frac{\pi}{6}$ or $\theta = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$.

- 14 Solve $2\cos^2 \theta - 1 = 0$ for θ in $[0, 2\pi)$.

$\cos^2 \theta = \frac{1}{2}$, so $\cos \theta = \pm \frac{1}{\sqrt{2}} = \pm \frac{\sqrt{2}}{2}$. Reference angle $\frac{\pi}{4}$, one solution in each quadrant:
 $\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$.

Angular and linear speed

- 15** A wheel of radius 10 cm rotates at 4 revolutions per second. Find its angular speed ω in radians per second, and the linear speed of a point on its rim.

$$\omega = 4 \times 2\pi = 8\pi \text{ rad/s (4 revolutions per second, each revolution} = 2\pi \text{ radians). Linear speed}$$
$$v = r\omega = 10(8\pi) = 80\pi \approx 251.3 \text{ cm/s.}$$

- 16** A satellite orbits Earth at a radius of 7000 km, completing one orbit every 2 hours. Find its linear speed in km/h.

$$\text{Angular speed } \omega = \frac{2\pi}{2} = \pi \text{ rad/h. Linear speed } v = r\omega = 7000\pi \approx 21,991 \text{ km/h.}$$

Right-triangle definitions and applications

- 17** A ladder leans against a wall, making a 65° angle with the ground. If the ladder is 12 ft long, find the height it reaches on the wall, to one decimal place.

$$\text{Height} = 12\sin(65^\circ) \approx 12(0.9063) \approx 10.9 \text{ ft.}$$

- 18** From a point 50 ft from the base of a tower, the angle of elevation to the top is 40° . Find the height of the tower, to one decimal place.

$$\text{Height} = 50\tan(40^\circ) \approx 50(0.8391) \approx 42.0 \text{ ft.}$$

- 19** A kite string is 80 ft long and makes a 55° angle with the ground. Find the height of the kite above the ground, and the horizontal distance from the flyer to a point directly below the kite, both to one decimal place.

$$\text{Height} = 80\sin(55^\circ) \approx 80(0.8192) \approx 65.5 \text{ ft. Horizontal distance}$$
$$= 80\cos(55^\circ) \approx 80(0.5736) \approx 45.9 \text{ ft.}$$

Multiple choice — easy 15

- 20** Find the exact value of $\sin(0)$.

a. $\frac{1}{2}$

b. 0 ✓

c. $\frac{\sqrt{2}}{2}$

d. $-\frac{\sqrt{2}}{2}$

$$\text{At } \theta = 0 (0^\circ): \sin(0) = 0.$$

21 Find the exact value of $\sin\left(\frac{7\pi}{6}\right)$.

a. 0

b. $\frac{\sqrt{3}}{2}$

c. $-\frac{1}{2}$ ✓

d. 1

At $\theta = \frac{7\pi}{6}$ (210°): $\sin\left(\frac{7\pi}{6}\right) = -\frac{1}{2}$.

22 Find the exact value of $\sin\left(\frac{5\pi}{4}\right)$.

a. $-\frac{\sqrt{3}}{2}$

b. -1

c. $-\frac{\sqrt{2}}{2}$ ✓

d. $\frac{\sqrt{3}}{2}$

At $\theta = \frac{5\pi}{4}$ (225°): $\sin\left(\frac{5\pi}{4}\right) = -\frac{\sqrt{2}}{2}$.

23 Find the exact value of $\cos\left(\frac{7\pi}{6}\right)$.

a. $-\frac{\sqrt{3}}{2}$ ✓

b. 0

c. -1

d. $-\frac{1}{2}$

At $\theta = \frac{7\pi}{6}$ (210°): $\cos\left(\frac{7\pi}{6}\right) = -\frac{\sqrt{3}}{2}$.

24 Find the exact value of $\cos(0)$.

a. 1 ✓

b. $\frac{\sqrt{2}}{2}$

c. $-\frac{\sqrt{3}}{2}$

d. $-\frac{1}{2}$

At $\theta = 0$ (0°): $\cos(0) = 1$.

25 Find the exact value of $\sin\left(\frac{2\pi}{3}\right)$.

a. $\frac{\sqrt{3}}{2}$ ✓

b. $-\frac{1}{2}$

c. 0

d. -1

At $\theta = \frac{2\pi}{3}$ (120°): $\sin\left(\frac{2\pi}{3}\right) = \frac{\sqrt{3}}{2}$.

26 Find the exact value of $\cos\left(\frac{11\pi}{6}\right)$.

a. $\frac{1}{2}$

b. 0

c. $\frac{\sqrt{3}}{2}$ ✓

d. $\frac{\sqrt{2}}{2}$

At $\theta = \frac{11\pi}{6}$ (330°): $\cos\left(\frac{11\pi}{6}\right) = \frac{\sqrt{3}}{2}$.

27 Find the exact value of $\sin\left(\frac{5\pi}{6}\right)$.

a. $\frac{1}{2}$ ✓

b. $-\frac{\sqrt{2}}{2}$

c. -1

d. 0

At $\theta = \frac{5\pi}{6}$ (150°): $\sin\left(\frac{5\pi}{6}\right) = \frac{1}{2}$.

28 Find the exact value of $\cos\left(\frac{3\pi}{4}\right)$.

a. 0

b. $\frac{1}{2}$

c. $-\frac{1}{2}$

d. $-\frac{\sqrt{2}}{2}$ ✓

At $\theta = \frac{3\pi}{4}$ (135°): $\cos\left(\frac{3\pi}{4}\right) = -\frac{\sqrt{2}}{2}$.

29 Find the exact value of $\sin\left(\frac{\pi}{3}\right)$.

a. $\frac{1}{2}$

b. $\frac{\sqrt{3}}{2}$ ✓

c. 1

d. $\frac{\sqrt{2}}{2}$

At $\theta = \frac{\pi}{3}$ (60°): $\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$.

30 Find the exact value of $\sin\left(\frac{5\pi}{3}\right)$.

a. $\frac{\sqrt{3}}{2}$

b. $-\frac{\sqrt{3}}{2}$ ✓

c. 1

d. -1

At $\theta = \frac{5\pi}{3}$ (300°): $\sin\left(\frac{5\pi}{3}\right) = -\frac{\sqrt{3}}{2}$.

31 Find the exact value of $\cos\left(\frac{\pi}{4}\right)$.

a. $\frac{1}{2}$

b. $-\frac{1}{2}$

c. 0

d. $\frac{\sqrt{2}}{2}$ ✓

At $\theta = \frac{\pi}{4}$ (45°): $\cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$.

32 Find the exact value of $\sin\left(\frac{\pi}{4}\right)$.

a. $-\frac{\sqrt{3}}{2}$

b. $\frac{\sqrt{2}}{2}$ ✓

c. $\frac{1}{2}$

d. 0

At $\theta = \frac{\pi}{4}$ (45°): $\sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$.

33 Find the exact value of $\cos\left(\frac{5\pi}{6}\right)$.

a. $-\frac{\sqrt{2}}{2}$

b. $-\frac{\sqrt{3}}{2}$ ✓

c. 0

d. -1

At $\theta = \frac{5\pi}{6}$ (150°): $\cos\left(\frac{5\pi}{6}\right) = -\frac{\sqrt{3}}{2}$.

34 Find the exact value of $\cos\left(\frac{4\pi}{3}\right)$.

a. $-\frac{1}{2}$ ✓

b. -1

c. $\frac{\sqrt{2}}{2}$

d. 0

At $\theta = \frac{4\pi}{3}$ (240°): $\cos\left(\frac{4\pi}{3}\right) = -\frac{1}{2}$.

Multiple choice — medium 15

35 Find the period of $f(x) = 3\sin(6x) - 4$.

a. $\frac{2\pi}{3}$

b. $\frac{\pi}{6}$

c. $\frac{\pi}{3}$ ✓

d. $\frac{4\pi}{3}$

$$\text{Period} = \frac{2\pi}{6} = \frac{\pi}{3}.$$

36 Find the period of $f(x) = 8\sin(5x) + 7$.

a. $\frac{4\pi}{5}$

b. $\frac{\pi}{5}$

c. $\frac{7\pi}{5}$

d. $\frac{2\pi}{5}$ ✓

$$\text{Period} = \frac{2\pi}{5} = \frac{2\pi}{5}.$$

37 Find the period of $f(x) = 9\cos(7x) - 8$.

a. $\frac{4\pi}{7}$

b. $\frac{\pi}{7}$

c. $\frac{2\pi}{7}$ ✓

d. $\frac{9\pi}{7}$

$$\text{Period} = \frac{2\pi}{7} = \frac{2\pi}{7}.$$

38 Find the period of $f(x) = 9\sin(5x) + 3$.

a. $\frac{7\pi}{5}$

b. $\frac{4\pi}{5}$

c. $\frac{2\pi}{5}$ ✓

d. $\frac{\pi}{5}$

$$\text{Period} = \frac{2\pi}{5} = \frac{2\pi}{5}.$$

39 Find the period of $f(x) = 5\sin(5x) - 2$.

a. $\frac{7\pi}{5}$

b. $\frac{4\pi}{5}$

c. $\frac{\pi}{5}$

d. $\frac{2\pi}{5}$ ✓

$$\text{Period} = \frac{2\pi}{5} = \frac{2\pi}{5}.$$

40 Find the period of $f(x) = 5\sin(6x)$.

a. $\frac{\pi}{3}$ ✓

b. $\frac{\pi}{6}$

c. $\frac{4\pi}{3}$

d. $\frac{2\pi}{3}$

$$\text{Period} = \frac{2\pi}{6} = \frac{\pi}{3}.$$

41 Find the period of $f(x) = 2\sin(4x) - 9$.

a. π

b. $\frac{\pi}{4}$

c. $\frac{\pi}{2}$ ✓

d. $\frac{3\pi}{2}$

$$\text{Period} = \frac{2\pi}{4} = \frac{\pi}{2}.$$

42 Find the period of $f(x) = 5\cos(4x) + 1$.

a. π

b. $\frac{\pi}{4}$

c. $\frac{3\pi}{2}$

d. $\frac{\pi}{2}$ ✓

$$\text{Period} = \frac{2\pi}{4} = \frac{\pi}{2}.$$

43 Find the period of $f(x) = 3\cos(4x) - 6$.

a. π

b. $\frac{\pi}{2}$ ✓

c. $\frac{3\pi}{2}$

d. $\frac{\pi}{4}$

$$\text{Period} = \frac{2\pi}{4} = \frac{\pi}{2}.$$

44 Find the period of $f(x) = 5\sin(4x) - 2$.

a. $\frac{\pi}{4}$

b. π

c. $\frac{\pi}{2}$ ✓

d. $\frac{3\pi}{2}$

$$\text{Period} = \frac{2\pi}{4} = \frac{\pi}{2}.$$

45 Find the period of $f(x) = 6\sin(8x) - 9$.

a. $\frac{5\pi}{4}$

b. $\frac{\pi}{2}$

c. $\frac{\pi}{8}$

d. $\frac{\pi}{4}$ ✓

$$\text{Period} = \frac{2\pi}{8} = \frac{\pi}{4}.$$

46 Find the period of $f(x) = 3\cos(3x) + 8$.

a. $\frac{5\pi}{3}$

b. $\frac{\pi}{3}$

c. $\frac{2\pi}{3}$ ✓

d. $\frac{4\pi}{3}$

$$\text{Period} = \frac{2\pi}{3} = \frac{2\pi}{3}.$$

47 Find the period of $f(x) = 9\cos(3x) + 8$.

a. $\frac{5\pi}{3}$

b. $\frac{\pi}{3}$

c. $\frac{4\pi}{3}$

d. $\frac{2\pi}{3}$ ✓

$$\text{Period} = \frac{2\pi}{3} = \frac{2\pi}{3}.$$

48 Find the period of $f(x) = 3\sin(3x) + 1$.

a. $\frac{4\pi}{3}$

b. $\frac{\pi}{3}$

c. $\frac{5\pi}{3}$

d. $\frac{2\pi}{3}$ ✓

$$\text{Period} = \frac{2\pi}{3} = \frac{2\pi}{3}.$$

49 Find the period of $f(x) = 2\sin(2x) + 8$.

- a. $\frac{\pi}{2}$
- b. 2π
- c. $\frac{3\pi}{2}$
- d. π ✓

$$\text{Period} = \frac{2\pi}{2} = \pi.$$

Multiple choice — hard 15

50 Given $\sin \theta = \frac{20}{29}$ with θ in Quadrant I, find $\cos \theta$.

- a. $\frac{21}{29}$ ✓
- b. $\frac{20}{29}$
- c. $-\frac{21}{29}$
- d. $\frac{29}{21}$

Using $\sin^2 \theta + \cos^2 \theta = 1$: $\cos^2 \theta = 1 - \left(\frac{20}{29}\right)^2 = \frac{441}{841}$, so $\cos \theta = \pm \frac{21}{29}$. In Quadrant I, cosine is positive: $\cos \theta = \frac{21}{29}$.

51 Given $\sin \theta = \frac{5}{13}$ with θ in Quadrant I, find $\cos \theta$.

- a. $\frac{12}{13}$ ✓
- b. $-\frac{12}{13}$
- c. $\frac{13}{12}$
- d. $\frac{5}{13}$

Using $\sin^2 \theta + \cos^2 \theta = 1$: $\cos^2 \theta = 1 - \left(\frac{5}{13}\right)^2 = \frac{144}{169}$, so $\cos \theta = \pm \frac{12}{13}$. In Quadrant I, cosine is positive: $\cos \theta = \frac{12}{13}$.

52 Given $\sin \theta = -\frac{20}{29}$ with θ in Quadrant IV, find $\cos \theta$.

a. $-\frac{20}{29}$

b. $-\frac{21}{29}$

c. $\frac{21}{29}$ ✓

d. $\frac{29}{21}$

Using $\sin^2 \theta + \cos^2 \theta = 1$: $\cos^2 \theta = 1 - \left(-\frac{20}{29}\right)^2 = \frac{441}{841}$, so $\cos \theta = \pm \frac{21}{29}$. In Quadrant IV, cosine is positive: $\cos \theta = \frac{21}{29}$.

53 Given $\sin \theta = \frac{8}{17}$ with θ in Quadrant I, find $\cos \theta$.

a. $-\frac{15}{17}$

b. $\frac{8}{17}$

c. $\frac{15}{17}$ ✓

d. $\frac{17}{15}$

Using $\sin^2 \theta + \cos^2 \theta = 1$: $\cos^2 \theta = 1 - \left(\frac{8}{17}\right)^2 = \frac{225}{289}$, so $\cos \theta = \pm \frac{15}{17}$. In Quadrant I, cosine is positive: $\cos \theta = \frac{15}{17}$.

54 Given $\sin \theta = \frac{3}{5}$ with θ in Quadrant I, find $\cos \theta$.

a. $\frac{5}{4}$

b. $\frac{4}{5}$ ✓

c. $\frac{3}{5}$

d. $-\frac{4}{5}$

Using $\sin^2 \theta + \cos^2 \theta = 1$: $\cos^2 \theta = 1 - \left(\frac{3}{5}\right)^2 = \frac{16}{25}$, so $\cos \theta = \pm \frac{4}{5}$. In Quadrant I, cosine is positive: $\cos \theta = \frac{4}{5}$.

55 Solve $\sin \theta - \left(-\frac{\sqrt{3}}{2}\right) = 0$ for θ in $[0, 2\pi)$.

a. $\theta = \frac{4\pi}{3}$ or $\theta = \frac{5\pi}{3}$ ✓

b. $\theta = \frac{\pi}{3}$ or $\theta = \frac{2\pi}{3}$

c. $\theta = \frac{7\pi}{6}$ or $\theta = \frac{11\pi}{6}$

d. $\theta = 0$ or $\theta = \pi$

$\sin \theta = -\frac{\sqrt{3}}{2}$. This occurs at $\theta = \frac{4\pi}{3}$ and $\theta = \frac{5\pi}{3}$.

56 Solve $2\cos \theta - 2\left(\frac{1}{2}\right) = 0$ for θ in $[0, 2\pi)$.

a. $\theta = \frac{\pi}{4}$ or $\theta = \frac{7\pi}{4}$

b. $\theta = \frac{3\pi}{4}$ or $\theta = \frac{5\pi}{4}$

c. $\theta = \frac{\pi}{3}$ or $\theta = \frac{5\pi}{3}$ ✓

d. $\theta = \frac{2\pi}{3}$ or $\theta = \frac{4\pi}{3}$

$\cos \theta = \frac{1}{2}$. This occurs at $\theta = \frac{\pi}{3}$ and $\theta = \frac{5\pi}{3}$.

57 Solve $4\cos \theta - 4\left(-\frac{1}{2}\right) = 0$ for θ in $[0, 2\pi)$.

a. $\theta = \frac{\pi}{2}$ or $\theta = \frac{3\pi}{2}$

b. $\theta = \frac{2\pi}{3}$ or $\theta = \frac{4\pi}{3}$ ✓

c. $\theta = \frac{5\pi}{6}$ or $\theta = \frac{7\pi}{6}$

d. $\theta = \frac{\pi}{3}$ or $\theta = \frac{5\pi}{3}$

$\cos \theta = -\frac{1}{2}$. This occurs at $\theta = \frac{2\pi}{3}$ and $\theta = \frac{4\pi}{3}$.

58 Solve $2\sin \theta - 2\left(\frac{\sqrt{3}}{2}\right) = 0$ for θ in $[0, 2\pi)$.

a. $\theta = \frac{\pi}{3}$ or $\theta = \frac{2\pi}{3}$ ✓

b. $\theta = \frac{\pi}{4}$ or $\theta = \frac{3\pi}{4}$

c. $\theta = \frac{4\pi}{3}$ or $\theta = \frac{5\pi}{3}$

d. $\theta = 0$ or $\theta = \pi$

$\sin \theta = \frac{\sqrt{3}}{2}$. This occurs at $\theta = \frac{\pi}{3}$ and $\theta = \frac{2\pi}{3}$.

59 Solve $\sin \theta - \left(-\frac{\sqrt{2}}{2}\right) = 0$ for θ in $[0, 2\pi)$.

a. $\theta = \frac{\pi}{3}$ or $\theta = \frac{2\pi}{3}$

b. $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$

c. $\theta = \frac{5\pi}{4}$ or $\theta = \frac{7\pi}{4}$ ✓

d. $\theta = \frac{\pi}{4}$ or $\theta = \frac{3\pi}{4}$

$\sin \theta = -\frac{\sqrt{2}}{2}$. This occurs at $\theta = \frac{5\pi}{4}$ and $\theta = \frac{7\pi}{4}$.

60 A ladder leans against a wall, making a 52° angle with the ground. If the ladder is 100 ft long, find the height it reaches on the wall, to one decimal place.

a. 78.8 ✓

b. 76.8

c. 80.8

d. 83.8

Height = $100\sin(52^\circ) \approx 100(0.788) \approx 78.8$ ft.

61 A ladder leans against a wall, making a 70° angle with the ground. If the ladder is 79 ft long, find the height it reaches on the wall, to one decimal place.

a. 76.2

b. 74.2 ✓

c. 79.2

d. 72.2

Height = $79\sin(70^\circ) \approx 79(0.9397) \approx 74.2$ ft.

- 62** A ladder leans against a wall, making a 38° angle with the ground. If the ladder is 26 ft long, find the height it reaches on the wall, to one decimal place.

a. 16.0 ✓

b. 14.0

c. 18.0

d. 21.0

$$\text{Height} = 26\sin(38^\circ) \approx 26(0.6157) \approx 16.0 \text{ ft.}$$

- 63** A ladder leans against a wall, making a 50° angle with the ground. If the ladder is 92 ft long, find the height it reaches on the wall, to one decimal place.

a. 75.5

b. 70.5 ✓

c. 72.5

d. 68.5

$$\text{Height} = 92\sin(50^\circ) \approx 92(0.766) \approx 70.5 \text{ ft.}$$

- 64** A ladder leans against a wall, making a 32° angle with the ground. If the ladder is 46 ft long, find the height it reaches on the wall, to one decimal place.

a. 24.4 ✓

b. 26.4

c. 22.4

d. 29.4

$$\text{Height} = 46\sin(32^\circ) \approx 46(0.5299) \approx 24.4 \text{ ft.}$$